

THREE STATEMENTS

Income Statement (I/S)

Revenue

-- COGS (Cost of Goods Sold)

Gross Profit

-- Operating Expenses

Operating Income (EBIT)

-- Interest Expense

+ Interest Income

Pretax Income

-- Taxes

Net Income

÷ Share Count

EPS

Balance Sheet (B/S)

Current Assets

Cash & Cash Equivalents

Short-term investments

Inventory

Accounts Receivables

Long Term Assets

PP&E

Total Assets

Current Liabilities

Short Term Debt

Accounts Payable

Long Term Liabilities

Long Term Debt

Total Liabilities

Common Stock & APIC

Retained Earnings

Total Shareholders Equity

Cash Flow Statement (CFS)

Operating Activities

Net Income

+ Non-cash Exp. (D&A, SBC)

+/- Dec./Inc. in Working Capital

Investing Activities

-- Capital Expenditures

Financing Activities

+ Cash from equity/debt financing

-- Share buyback/dividends

-- Debt repayment

= Net Change in Cash

Some common "links" for the three financial statements:

1) Net income on the I/S goes to the top of the CFS, and to retained earnings on the B/S

2) Non-cash expenses from the I/S need to get added back to net income on the CFS

3) Changes in operating working capital need to be added or subtracted under operating activities on the CFS

4) Capital Expenditures on the CFS adds to PP&E on the B/S, depreciation on the I/S decreases PP&E on the B/S

5) Increase or decrease in equity under financing activities on the CFS affects shareholders equity on the B/S

6) Increase or decrease in debt under financing activities affects debt under liabilities on the B/S

7) Net change in cash on the CFS (sum of changes across all 3 sections) impacts cash on the B/S

EQUITY VALUE AND ENTERPRISE VALUE

Equity Value (a.k.a. "Market Cap") =
Share Price x Number of Shares

Enterprise Value (a.k.a. "EV") = Equity
Value - Cash & Cash Equivalents +
Debt + Preferred + Minority Interest

Diluted Share Count = Basic Share
Count + Additional Shares Issued
From "Dilutive Securities"

Dilutive Securities include stock
options, warrants, restricted stock
units, convertible debt, convertible
preferred stock, etc.

Use "Treasury Stock Method" to
calculate diluted shares when it comes
to Stock Options and Warrants. This
means only "in-the-money" securities
are converted to shares, and the
exercise proceeds are used to buy
back as many shares as possible

Common valuation multiples:

EV/Revenue, EV/EBITDA, EV/EBIT,
and P/E (same as Market Cap / Net
Income).

There are also other less common
multiples that are industry-specific

Note: Enterprise Value must be used if
the metric being compared is above
the Interest Expense line on the
Income Statement. Equity Value is
used for metrics below the Interest
Expense line ("apples to apples"
comparison)

VALUATION METHODOLOGIES

Comparable Company (a.k.a. "Public Comps" or "Trading Comps")

1) Look at other public companies that
are similar in terms of industry /
business model, size / financial profile,
and sometimes geography

2) Calculate the appropriate valuation
multiple for the comp set

3) Justify what multiple the company
you're trying to value should have
relative to the range of multiples you
got from the comp set. Typically,
companies with higher growth /
margins deserve higher multiples, and
vice versa

Precedent Transactions (a.k.a. "M&A Comps" or "Transaction Comps")

1) Look at other companies that are
similar in terms of industry / business
model, size / financial profile, and
sometimes geography, and have been
acquired recently (during similar
market conditions)

2) Calculate the valuation multiple they
got acquired at, including the control
premium that was paid

3) Justify what multiple the company
you're trying to value should have
relative to the range of multiples you
got from the comp set. Typically,
companies with higher growth /
margins deserve higher multiples, and
vice versa

Discounted Cash Flow (a.k.a. "DCF")

1) Concept: a company is worth the
sum of its future cash flows,
discounted back to present value

2) Calculate Unlevered Free Cash Flow for the next 5-10 years: $EBIT * (1 - \text{Tax}) + \text{Non-Cash Expenses} \pm \text{Changes in Operating Working Capital} - \text{Capital Expenditures}$

3) Calculate the Discount Rate, or Weighted Average Cost of Capital (a.k.a. "WACC"): $\text{Cost of Equity} * \% \text{Equity} + \text{Cost of Debt} * \% \text{Debt} * (1 - \text{Tax Rate}) + \text{Cost of Preferred} * \% \text{Preferred Stock}$

4) Calculate Cost of Equity using CAPM: $\text{Risk-Free Rate} + \text{Levered Beta} * \text{Equity Risk Premium}$

Risk-Free Rate = US T-Bill Yield

Unlevered Beta = Levered Beta / $(1 + (1 - \text{Tax Rate}) * (\text{Debt} / \text{Equity}))$

Levered Beta = Unlevered Beta * $(1 + (1 - \text{Tax Rate}) * (\text{Debt} / \text{Equity}))$

Equity Risk Premium = Stock Market Returns - Risk Free Rate

5) Calculate Terminal Value:

Exit Multiple Method: take the EBITDA in the last year of the forecast period and multiply it by an appropriate EV/EBITDA multiple based on the comps

Gordon Growth Method: $\text{Final Year FCF} * (1 + \text{Terminal FCF Growth Rate}) / (\text{Discount Rate} - \text{Terminal FCF Growth Rate})$

6) Discount all forecasted FCF and terminal value using the discount rate to get to the enterprise value

[Note] If using Levered FCF for DCF, discount using cost of equity instead of WACC, and derive equity instead of enterprise value

MERGERS AND ACQUISITIONS (ACCRETION / DILUTION)

When AcquireCo acquires TargetCo, AcquireCo's investors want to know that the deal will bring more profits to the company than what it costs. The easiest way to think about this is to see if the earnings per share (EPS) increases or decreases after the transaction is done

A deal is accretive when the post-transaction EPS is higher than the pre-transaction EPS

A deal is dilutive when the post-transaction EPS is lower than the pre-transaction EPS (bad)

To calculate whether a deal is accretive or dilutive, you basically take the income statement of both companies and combine them, line by line. For example:

$$\text{AcquireCo revenue} + \text{TargetCo revenue} = \text{CombinedCo revenue}...$$

$$\text{AcquireCo COGS} + \text{TargetCo COGS} = \text{CombinedCo COGS}...$$

And so on so forth.

Some adjustments to be aware of:

1) Factor in any potential revenue synergies and cost synergies:

Revenue synergies are things like price increases, or upsell and cross-sell opportunities

Cost synergies are things like headcount reduction, where the companies consolidate their teams and let go of redundant headcount

2) Adjust for the cost of doing the deal, which is dependent on:

The purchase price, or how much AcquireCo has to pay, and

The purchase method, or whether AcquireCo is paying with cash, debt, stock, or a combo of the above. Each method has differing costs, from lowest to highest as follows:

Cash: foregone interest income

Debt: interest expense incurred

Equity: increase in share count; lowers EPS because existing investors have to share future upside with new investors

In the end, the accretion / dilution

calculation is simple:

IF $\text{TargetCo Net Income} + \text{Revenue}$ and $\text{Cost Synergies} > \text{Cost of Doing Deal}$, it's accretive

IF $\text{TargetCo Net Income} + \text{Revenue}$ and $\text{Cost Synergies} < \text{Cost of Doing Deal}$, it's dilutive

LEVERAGED BUYOUTS (LBO)

A LBO is when a private equity firm buys a company using both debt and cash they raised, and then sells it in 3-5 years for a profit

Three ways to increase the IRR:

1) How much the PE firm pays: buying at a lower entry multiple, or selling at a higher exit multiple

2) How the PE firm pays: what's the max debt load without going bankrupt? More debt = higher IRR

3) How the company performs: the more free cash flow it generates, the more debt gets paid down, which means more equity upon exit for the PE firm

Qualities of good LBO candidates:

Stable and predictable cash flow, undervalued, strong management, cost-cutting opportunities, low capex, and strong collateral base

Steps of the LBO Model:

- 1) Determine purchase price
- 2) Determine amount and type of debt used, plus interest and terms
- 3) Create sources and uses
- 4) Project out the financial statements and calculate FCF
- 5) Use FCF to pay off as much of the debt as possible
- 6) Determine exit price, and calculate IRR: $\text{MOIC}^{(1/\text{years})} - 1$

[Note] Typically you don't need to mention this as a valuation methodology during interviews, but if they ask you for a fourth one you would use an LBO. You basically figure out the highest price a private equity firm would be willing to pay to still be able to hit at least 15-20% IRR in the deal